

Python, FastAPI & Pytest Master Program

LEVEL 1: Fundamentals of Python to Advanced

Module 1: Introduction to Python

- What is Python?
- Features of Python
- Python Installation
- VS Code Setup
- Running Python Programs
- Python File Structure
- Virtual Environment (venv) and pip

Module 2: Variables & Data Types

- Variables
- Naming Conventions
- Data Types
- Type Casting
- Input & Output

Module 3: Operators

- Arithmetic Operators
- Assignment Operators
- Comparison Operators
- Logical Operators
- Identity Operators (`is`, `is not`)
- Membership Operators (`in`, `not in`)
- Operator Precedence

Module 4: Conditional Statements

- `if`
- `if...else`
- `if...elif...else`
- Nested `if`
- Match Case
- Ternary Operator

Module 5: Loops

- `for` Loop
- `while` Loop



- `range()`
- `enumerate()`
- `break`
- `continue`
- `pass`
- Nested Loops

Module 6: Strings

- Indexing
- Slicing
- String Methods
- Formatting [`split()` , `join()` , `replace()` , `find()`]

Module 7: Lists

- List Basics
- List Methods
- Nested Lists
- List Comprehension
- Sorting
- Searching

Module 8: Tuples & Sets

- Tuple Basics
- Tuple Methods
- Packing & Unpacking
- Set Basics
- Set Methods
- Union
- Intersection
- Difference

Module 9: Dictionary

- Dictionary Basics
- CRUD Operations
- Nested Dictionary
- Dictionary Methods
- Dictionary Comprehension

Module 10: Functions

- Functions
- Parameters



- Arguments
- Return
- Default Arguments
- Keyword Arguments
- Lambda

Module 11: OOP

- Class
- Object
- Constructor (`__init__`)
- Instance Variables
- Methods
- Encapsulation
- Inheritance
- Polymorphism
- Abstraction

Module 12: Exception Handling

- `try`
- `except`
- `finally`
- `raise`
- Custom Exception

Module 13: File Handling

- Text Files
- JSON Files
- CSV Files
- Read & Write Files

Module 14: JSON

- JSON Basics [`load()`, `loads()`, `dump()`, `dumps()`]
- Nested JSON
- JSON Parsing

Module 15: Advanced Python

- List Comprehension
- Dictionary Comprehension
- `map()`
- `filter()`
- Iterators



- Generators
- Decorators

LEVEL 2: Fast API

Module 1: Introduction to FastAPI

- What is FastAPI?
- Why FastAPI?
- Installation
- VS Code Setup
- Project Structure
- Uvicorn
- Swagger UI

Module 2: API Basics

- HTTP Methods
- GET
- POST
- PUT
- PATCH
- DELETE
- Status Codes

Module 3: Routing

- Path Parameters
- Query Parameters
- Route Parameters
- Request & Response

Module 4: Pydantic Models

- BaseModel
- Request Body
- Response Model
- Field Validation
- Optional Fields

Module 5: CRUD Operations

- Create
- Read
- Update
- Delete



- Simple CRUD Project

Module 6: Authentication

Module 7: Exception Handling

- HTTP Exception
- Custom Exceptions
- Validation Errors

Module 8: Middleware

- Middleware
- CORS
- Logging

Module 9: Dependency Injection

- Depends()
- Reusable Dependencies

Module 10: File Handling

- File Upload
- File Download
- Static Files

Module 11: Project Structure

- Routers
- Models
- Schemas
- Services
- Utils
- Config



LEVEL 3: Pytest

Module 1: Introduction

- What is Pytest?
- Why Pytest?
- Installation
- Project Setup
- First Test
- Running Tests

Module 2: Test Structure

- Test Naming Convention
- Test Functions
- Assertions
- Test Discovery

Module 3: Fixtures

- What are Fixtures?
- Creating Fixtures
- Fixture Scope
- `conftest.py`

Module 4: Parameterization

- `@pytest.mark.parametrize`
- Data-Driven Testing

Module 5: Markers

- Smoke
- Sanity
- Regression
- Skip
- xfail

Module 6: Reports

- HTML Report
- Test Summary

